🎬 Netflix Data Analysis Project

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1. 📜 Project Overview

This project focuses on analyzing Netflix's content dataset to uncover insights into content distribution, growth trends, genres, and country-wise production.

The objective is to:

- Understand the distribution of Movies vs TV Shows
- Identify top-performing countries and genres
- Analyze content growth over time
- Explore business insights using SQL queries
- Visualize findings through Power BI dashboard

This project demonstrates **end-to-end data analysis skills** including Python, SQL, and data visualization.

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2. 📂 Dataset Description

- **Total Records:** 8,807
- **Total Columns:** 12

Key Features:

- Show ID, Title, Type (Movie / TV Show)
- Director, Cast
- Country
- Date Added
- Release Year
- Rating
- Duration
- Genre (listed_in)
- Description

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train I...

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3. 🧹 Data Cleaning & Preparation (Python)

Data preprocessing was performed using Python libraries like **Pandas, NumPy, Matplotlib, and Seaborn**.

Key Steps:

- Loaded dataset using pandas.read_csv()
- Explored structure using .info() and .shape()
- Checked missing values

Data Cleaning:

- Filled missing values:
  - country → "Unknown"
  - director → "Unknown"
- Converted date_added to datetime format
- Removed duplicates

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   show_id       8807 non-null   object
 1   type          8807 non-null   object
 2   title         8807 non-null   object
 3   director      6173 non-null   object
 4   cast          7982 non-null   object
 5   country       7976 non-null   object
 6   date_added    8797 non-null   object
 7   release_year  8807 non-null   int64
 8   rating        8803 non-null   object
 9   duration      8804 non-null   object
 10  listed_in     8807 non-null   object
 11  description   8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

2.1 check missing values

```
df.isnull().sum()
```

```
show_id          0
type             0
title            0
director      2634
cast           825
country        831
date_added      10
release_year     0
rating           4
duration         3
listed_in        0
description      0
dtype: int64
```

Feature Engineering:

- Split listed_in column into **individual genres**
- Expanded dataset using .explode() for better analysis

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4. 📊 Exploratory Data Analysis (Python)

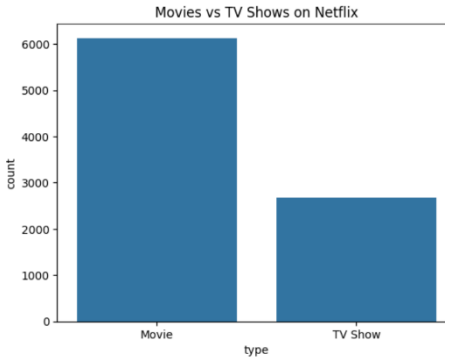
Key Analysis Performed:

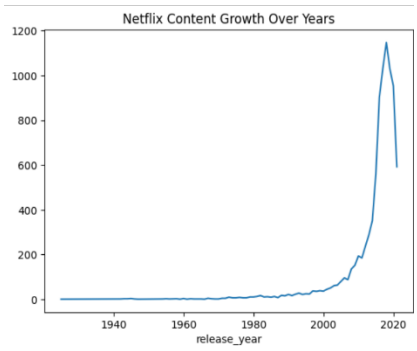
- Distribution of **Movies vs TV Shows**
- Content growth over years
- Top 10 countries producing content
- Most popular genres

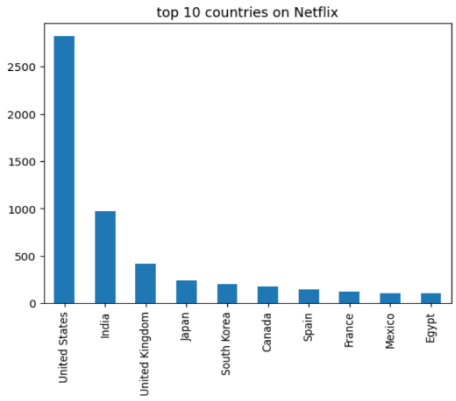
Visualization:

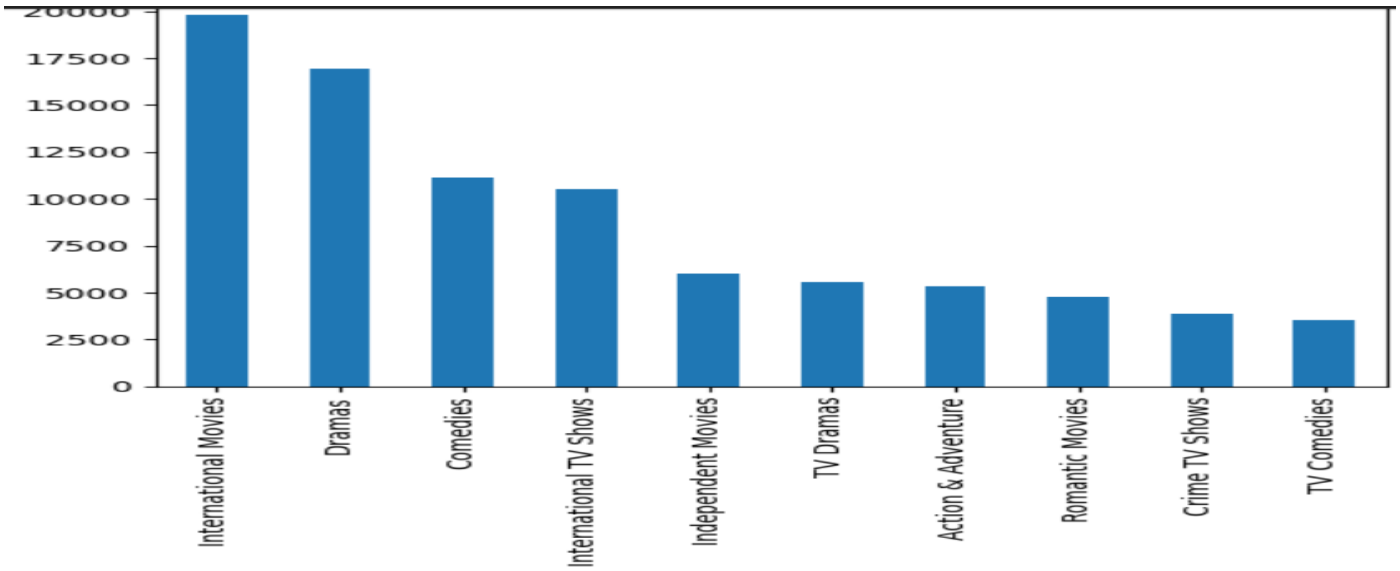
- Bar charts for country & genre analysis

- Line chart for content growth
- Count plots for content type









Top genres on Netflix

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5. 🧠 Recommendation System (Python)

A simple content-based recommendation function was created:

- Recommends titles based on **same genre**
- Helps users discover similar content

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6. 🗄️ SQL Analysis (Business Insights)

SQL was used to perform advanced business analysis on cleaned data.

Key Insights:

- Total number of titles available

```sql
-- Count total number of titles in the dataset.
select count(distinct show_id) as total_titles
from  final_netflix  ;
```

- Distribution of Movies vs TV Shows

```sql
-- Find how many Movies vs TV Shows are available.
select type , count(*) as total_count
 from final_netflix
 group by type ;
```

- Top 5 countries with highest content

```sql
-- List the top 5 countries with the most content.
select country ,count(*) as total_content
 from  final_netflix
 where country  is not null
  group by country
  order by count(*) desc
  limit 5 ;
```

- Most common content ratings

```sql
-- Find the most common rating (e.g., TV-MA, PG).
select rating , count(*) as total_rating
from final_netflix
 group by  rating
 order by count(*) desc ;
```

Business-Level Queries:

- High-performing genres

```sql
-- Which genres have the highest number of titles?
select genres ,count(*) as total_genres
from final_netflix
group by genres
 order by count(*) desc
  limit 1 ;
```

- Recent content additions

```sql
-- Find the top 10 most recent content additions.
select   distinct title ,  type ,date_format(date_added, '%Y -%m') as date_added
from final_netflix
where date_added is not null
  order by date_added desc
limit 10 ;
```

- Country with most TV Shows

```sql
-- Which country produces the most TV Shows only?
 select  distinct country  , count(*) as total_tv_shows
 from final_netflix
where type = 'TV Show'
group by country
 order by count(*)  desc
 limit 1 ;
```

- Average release year

```sql
-- Find the average release year of content on the platform
select round(ifnull(avg(release_year),0),2) as avg_release_year
from final_netflix  ;
```

Advanced SQL Analysis:

- Year-wise content growth

```sql
-- Find the year-wise growth of content added to the platform.
WITH cte AS (
  SELECT
    YEAR(STR_TO_DATE(date_added, '%M %d, %Y')) AS year_added,
    COUNT(*) AS total_titles
  FROM final_netflix
  WHERE date_added IS NOT NULL
  GROUP BY year_added
)
SELECT
  year_added,
  total_titles,
  total_titles - LAG(total_titles) OVER (ORDER BY year_added) AS growth
FROM cte;
```

- Country-wise growth trends

```sql
-- Identify countries with increasing content trend over time.
 WITH yearly_data AS (
  SELECT
    country,
    YEAR(STR_TO_DATE(date_added, '%M %d, %Y')) AS year_added,
    COUNT(*) AS total_titles
  FROM final_netflix
  WHERE date_added IS NOT NULL
    AND country IS NOT NULL
  GROUP BY country, year_added
),
growth_data AS (
  SELECT
    country,
    year_added,
    total_titles,
    total_titles - LAG(total_titles) OVER (PARTITION BY country ORDER BY year_added) AS growth
  FROM yearly_data
)
SELECT DISTINCT country
FROM growth_data
WHERE growth > 0;
```

- Most popular genre per country

```sql
-- Which genre is most popular in each country?
WITH RECURSIVE split_data AS (
 SELECT
  show_id,
  TRIM(SUBSTRING_INDEX(country, ',', 1)) AS country,
  SUBSTRING(country, LENGTH(SUBSTRING_INDEX(country, ',', 1)) + 2) AS rest_country,
  TRIM(SUBSTRING_INDEX(listed_in, ',', 1)) AS genre,
  SUBSTRING(listed_in, LENGTH(SUBSTRING_INDEX(listed_in, ',', 1)) + 2) AS rest_genre
 FROM final_netflix

 UNION ALL

 SELECT
  show_id,
  TRIM(SUBSTRING_INDEX(rest_country, ',', 1)),
  SUBSTRING(rest_country, LENGTH(SUBSTRING_INDEX(rest_country, ',', 1)) + 2),
  TRIM(SUBSTRING_INDEX(rest_genre, ',', 1)),
  SUBSTRING(rest_genre, LENGTH(SUBSTRING_INDEX(rest_genre, ',', 1)) + 2)
 FROM split_data
 WHERE rest_country <> '' OR rest_genre <> ''
),

counts AS (
 SELECT
  country,
  genre,
  COUNT(*) AS total
 FROM split_data
 WHERE country <> '' AND genre <> ''
 GROUP BY country, genre
),

ranked AS (
 SELECT *,
```

```
    RANK() OVER (PARTITION BY country ORDER BY total DESC) AS rnk
  FROM counts
)

SELECT country, genre, total
FROM ranked
WHERE rnk = 1;
```

- Top contributing genres (50% content share)

```
-- Find the top 3 genres contributing to 50% of total content
WITH genre_counts AS (
  SELECT
    listed_in AS genre,
    COUNT(*) AS total
  FROM final_netflix
  GROUP BY listed_in
),
total_count AS (
  SELECT SUM(total) AS grand_total FROM genre_counts
),
ranked AS (
  SELECT
    g.genre,
    g.total,
    SUM(g.total) OVER (ORDER BY g.total DESC) AS running_total,
    t.grand_total
  FROM genre_counts g
  CROSS JOIN total_count t
)
SELECT
  genre,
  total,
  ROUND((running_total / grand_total) * 100, 2) AS cumulative_percentage
FROM ranked
WHERE (running_total / grand_total) <= 0.5
limit 3 ;
```

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7. 📊 Power BI Dashboard

An interactive dashboard was created to present insights visually.

Dashboard Highlights:

- Content distribution (Movies vs TV Shows)
- Country-wise content production
- Genre popularity
- Year-wise growth trends
- KPI metrics for quick insights

This dashboard helps stakeholders make **quick and data-driven decisions**.

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8. 💡 Key Insights

- Movies dominate Netflix content compared to TV Shows
- Certain countries contribute significantly more content

- Content production increased rapidly after 2015
- Drama and International genres are highly popular
- Netflix is continuously expanding its global content library



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9. 📌 Business Recommendations

🔹 Content Strategy

- Focus on high-performing genres like Drama & International content

🔹 Global Expansion

- Invest more in top-performing countries

🔹 User Engagement

- Improve recommendation systems

🔹 Content Planning

- Continue increasing content production based on trends

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10. 🚀 Tools & Technologies Used

- **Python** (Pandas, NumPy, Matplotlib, Seaborn)

- **SQL** (MySQL)
- **Power BI** (Dashboard & Visualization)
- **GitHub** (Project Hosting)

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11. 📚 Key Learnings

- Data cleaning and preprocessing techniques
- Writing efficient SQL queries
- Data visualization and storytelling
- Building end-to-end analytics projects

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12. 🔮 Future Improvements

- Build advanced recommendation system
- Deploy dashboard online
- Use machine learning for prediction
- Automate data pipeline